GREEN ALIBI

Testing Whether Fluorescence Catches Drought Stress Before the Eye Can See It

Executive Summary · DOI: 10.5281/zenodo.21762501 · Sakshi D. Maske

The Question

India's official drought-declaration process, and the PMFBY crop-insurance payout mechanism that depends on it, rely substantially on NDVI — a reflectance signal that only changes once a plant's structure has already begun to visibly degrade. Solar-Induced Fluorescence (SIF) is grounded in a more direct physical process: a drop in photosynthetic efficiency, before outward greenness changes. Does that physical head-start actually show up as a measurable time lag on real satellite data — and does drought severity make that lag bigger, as hypothesized?

The Method

Eight districts of Marathwada, Maharashtra were tested across two drought years (2015, 2018) and one normal monsoon year (2020), using GOSIF v2 fluorescence data and cloud-screened MODIS NDVI. The SIF-to-NDVI decline lag was calculated two independent ways — threshold-crossing (linear-interpolation at 5 seasonal decline thresholds) and time-lagged cross-correlation — with 2,000-replicate bootstrap confidence intervals quantifying precision. Drought years were independently confirmed against CHIRPS rainfall data and a real government drought-declaration date.

The Finding

SIF consistently declines before NDVI in all three years — confirmed by two independent methods, and robust in direction under bootstrap resampling (SIF's lag was non-negative in 100% of replicates, every year). But drought severity does not amplify the lag: the two drought years produced a smaller average lag (15.6 days) than the normal year (28.7 days) — the opposite of the predicted direction.

Year	Threshold-Crossing Lag (mean)	Cross-Correlation Lag (95% CI)
2015 (drought)	24.3 days	13 days [5, 20]
2018 (drought)	6.9 days	4 days [0, 9]
2020 (normal)	28.7 days	7 days [0, 32]

H1 (SIF leads NDVI) is supported by both methods. H3 (drought amplifies the lag) is explicitly reported as not supported — a genuine, honestly-disclosed non-result, not a confirmed contrary finding, given the small sample (n=3 years).

Validation & Robustness Checklist

- ✓ Two independent lag-estimation methods (threshold-crossing + cross-correlation)
- ✓ Bootstrap confidence intervals (2,000 replicates per year)
- ✓ Independent rainfall validation (CHIRPS, 20-year climatological baseline)
- ✓ Moran's I spatial-autocorrelation check (effective sample size disclosed)
- ✓ Cross-referenced against the real government drought-declaration date (2018)
- ! H3 (drought amplifies lag) — honestly reported as not supported
- ! Exact year-to-year ranking flagged as method-sensitive, not robust

Honest Limitation

Every lag value is a point estimate from only 17 post-peak satellite observations per year — confidence intervals are wide, and every pairwise comparison between years' intervals overlaps, so the exact year-to-year ranking is not statistically distinguishable from noise at this sample size. The SIF-rainfall spatial correlation (Pearson r=0.837, p<0.001, n=24 district-years) is genuine, but a Moran's I diagnostic confirms the effective independent sample size is closer to 3 (one per study year) than 24, since neighboring districts share regional weather systems.

Real-World Relevance

Comparing 2018's SIF and NDVI decline onset against Maharashtra's official drought declaration (31 October 2018) shows both satellite indicators leading the official declaration by roughly seven weeks — with SIF's own edge over NDVI a further three to four days on top of that. For a rainfall-deficit-prone farming region, an earlier, physically-grounded signal feeding into drought declaration and PMFBY crop-insurance timing is the difference between relief that arrives in time and relief that arrives after the season is already lost.

GitHub: github.com/sakshimaske303-commits/GREEN_ALIBI | **Live Dashboard:** greenalibi-bzs2wvod5fflqh7dfe2cf4.streamlit.app | **Zenodo DOI:** 10.5281/zenodo.21762501

Sakshi D. Maske — Independent Geospatial Researcher